<u>Applications of Artificial Intelligence</u>

<u>Unit 1 - Consolidate Task</u>

This task is designed to help you evaluate and solidify your comprehension of fundamental AI concepts and terminology. Try to answer as concisely as you can -- each question's answer should be no more than 50 words in total.

<u>Responses</u>

1. **Define AI. How does it differ from human intelligence?**

   AI is the application of computer systems to complete tasks or create output that traditionally requires human intervention. It possesses sophisticated decision-making and reasoning ability that stimulates human intelligence.

   AI is powered by data and pattern recognition, while human intelligence is natural, based on intuition and experience.

2. **Describe the significance of the Turing Test and why it is still relevant today.**

   The Turing Test was proposed by Alan Turing in the 1950s.
   The test examines whether a human can distinguish between human-generated and computer-generated answers to prompts.
   The Turing Test is relevant today as it can be used to measure the capability for "intelligence" possessed by modern AI systems.

3. **Explain the difference between weak AI and strong AI and provide examples of each.**

   Weak AI (Narrow AI) performs specific tasks. It does not truly think but works and reasons through training, data and prompts. Examples are personal assistants (like Siri) and chess engines.

   Strong AI (General AI) performs a wider range of tasks, learning and possessing self-awareness. It is generally hypothetical, as it does not exist yet.

4. **What is neuro-symbolic AI? Why is it considered a hybrid approach?**

   Neuro-symbolic AI combines intelligence from **data from neural networks** (such as from deep-learning by working through big data) and **symbolic AI** (which is rule-based).

Neural networks represent the ability to learn, while symbolic AI represents the ability to reason. Neuro-symbolic AI combines the two.

5. **List three historical milestones in AI development and their impact.**
   - **1950** – The **Turing Test** formulated through the Imitation Game in Alan Turing's paper "Computing Machinery and Intelligence"
   - **1997** – IBM's **Deep Blue defeats world chess champion Kasparov**. This demonstrates the machine's ability to reason similarly to humans.
   - **2022** – Launch of **OpenAI's ChatGPT** brings generative AI into mainstream use.

6. **Describe the role of System 1 and System 2 thinking in human intelligence. How might these systems influence AI design?**

   Introduced by **Kahneman (2011)**, **System 1** thinking is **low-level, intuitive,** and can be done on instinct (e.g. humans walking, solving basic maths, and recognising people).

   System 2 thinking is **higher-level and logical**, requiring deliberation (e.g. humans playing chess or solving complex maths).

   Current AI systems perform most processes with their version of system 1 thinking.

7. **What are the strengths and limitations of AI in predictive policing**?

   Strengths:
   - **Conserves public funds** by ensuring police are present only where they are most likely to be needed.

   Limitations
   - **Reinforces bias** as it is based on past data.
   - It **might not account for societal changes,** improvements in living standards and employment, which can radically change the nature of crime in an area.

8. **Define skill-acquisition efficiency in the context of AI.**

   Skill acquisition efficiency is the efficiency at which an entity can learn new skills, considering the inputs in training the entity, such as cost, time, and resource investment. Compared to humans, AI can generally be taught new skills quickly.

9. **Identify two current applications of generative AI and discuss their societal impacts.**

   Education: Generative AI can be used to produce learning resources (like notes and practice tests) customised to individual student needs. However, such materials can contain outdated information and inaccuracies.

   Gaming: Including stimulated opponents in computer games like chess, users can select the level of expertise of the stimulated opponent. However, these gaming experiences can lack the benefits of natural human interaction.

10. **What is the relationship between data availability and the development of AI systems?**

    Data availability supports the development of AI systems and determines which problems AI systems can solve.
    AI domains with high data availability can advance quickly. (for example, chatbots in popular languages, systems for chess and science).
    AI in domains with lower data availability will advance slower (for example, chatbots in less popular languages, and systems for detecting rare diseases).

Draft Bibliography

1. Britannica. (n.d.). *Turing test*. Encyclopedia Britannica. https://www.britannica.com/technology/Turing-test

2. IBM. (n.d.). *What is strong AI?* IBM Think. https://www.ibm.com/think/topics/strong-ai